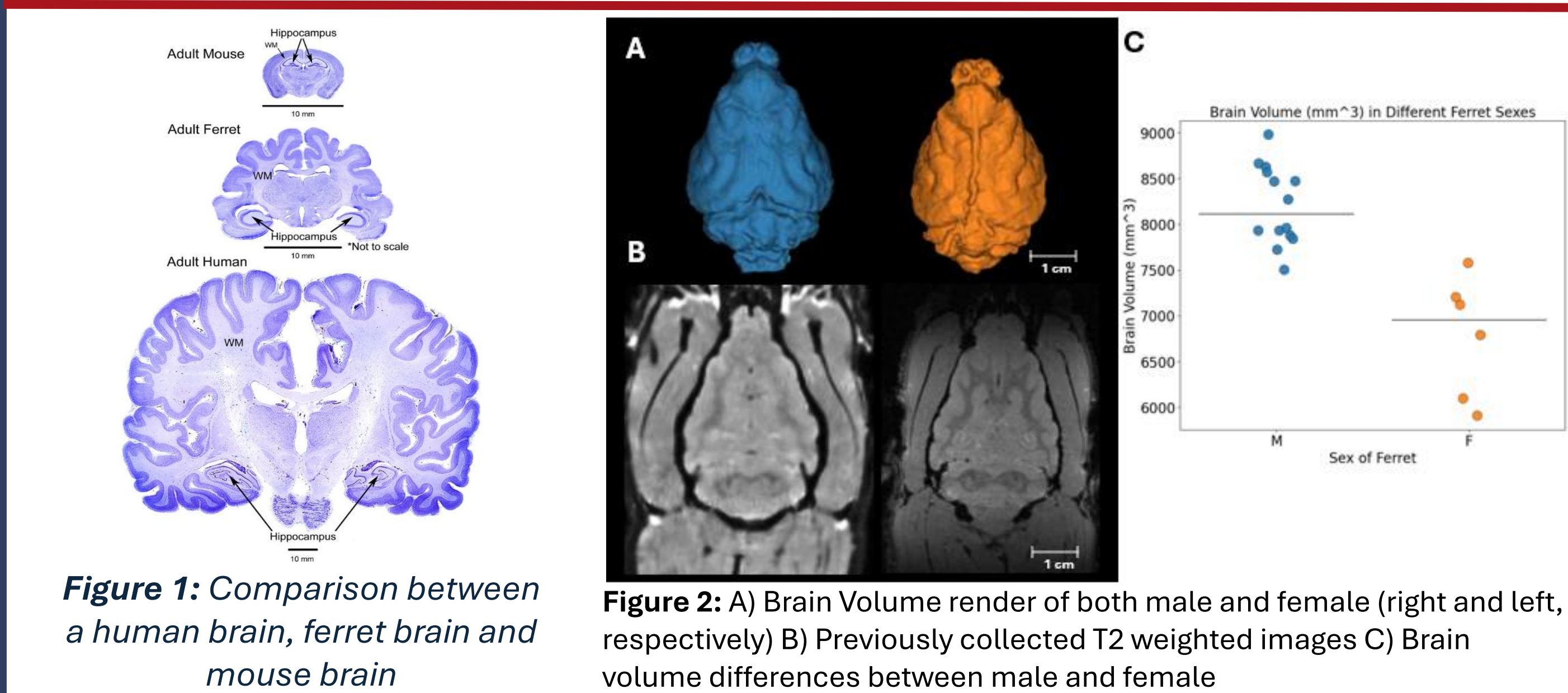


Development of 3D Printed Ferret Cradle with Anesthesia and Commercial Coil Integration for Brain MRI

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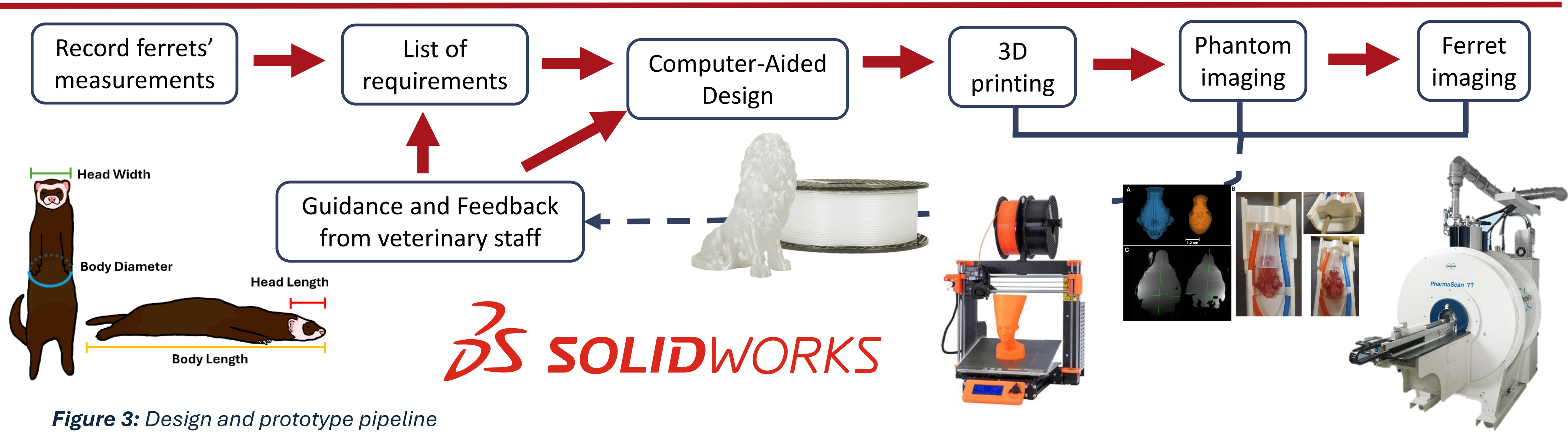
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Introduction



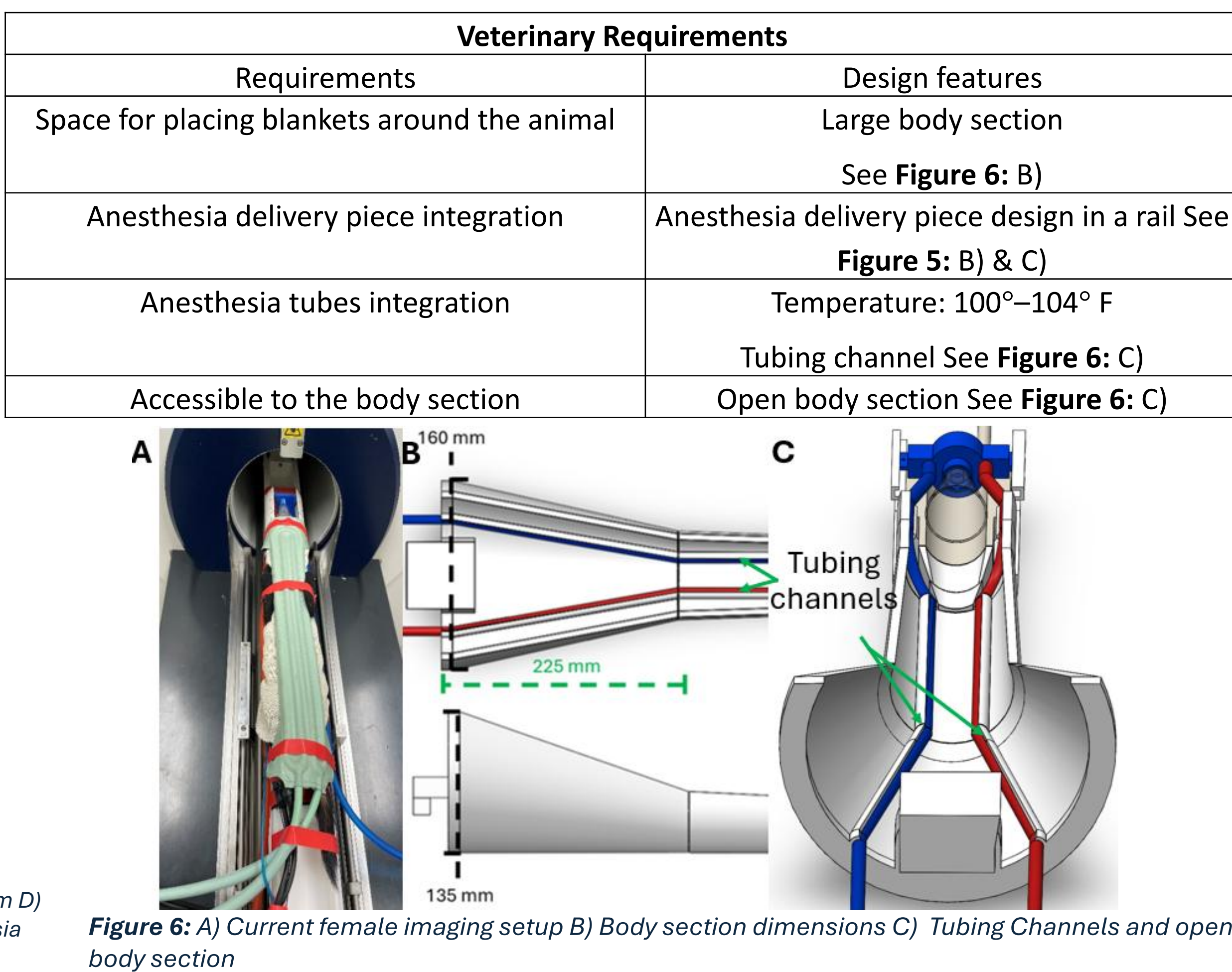
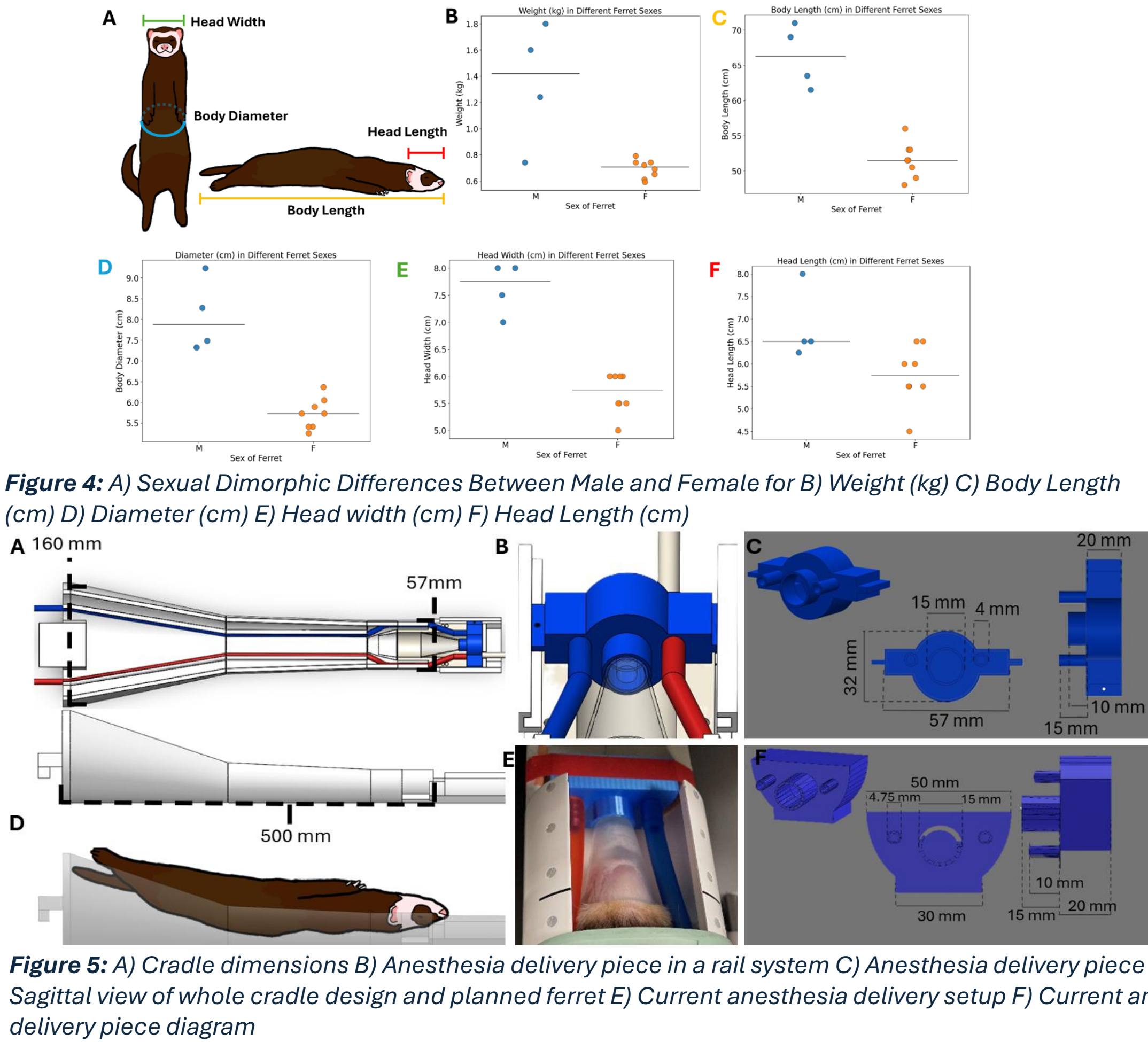
- Ferrets are a promising pre-clinical model for brain MRI studies
- Sexual dimorphic differences in ferrets
- There is no a commercial solution
- Improvements to the current setup for female imaging based on guidance from veterinary staff
- 3D-printed custom-made solutions

Methods

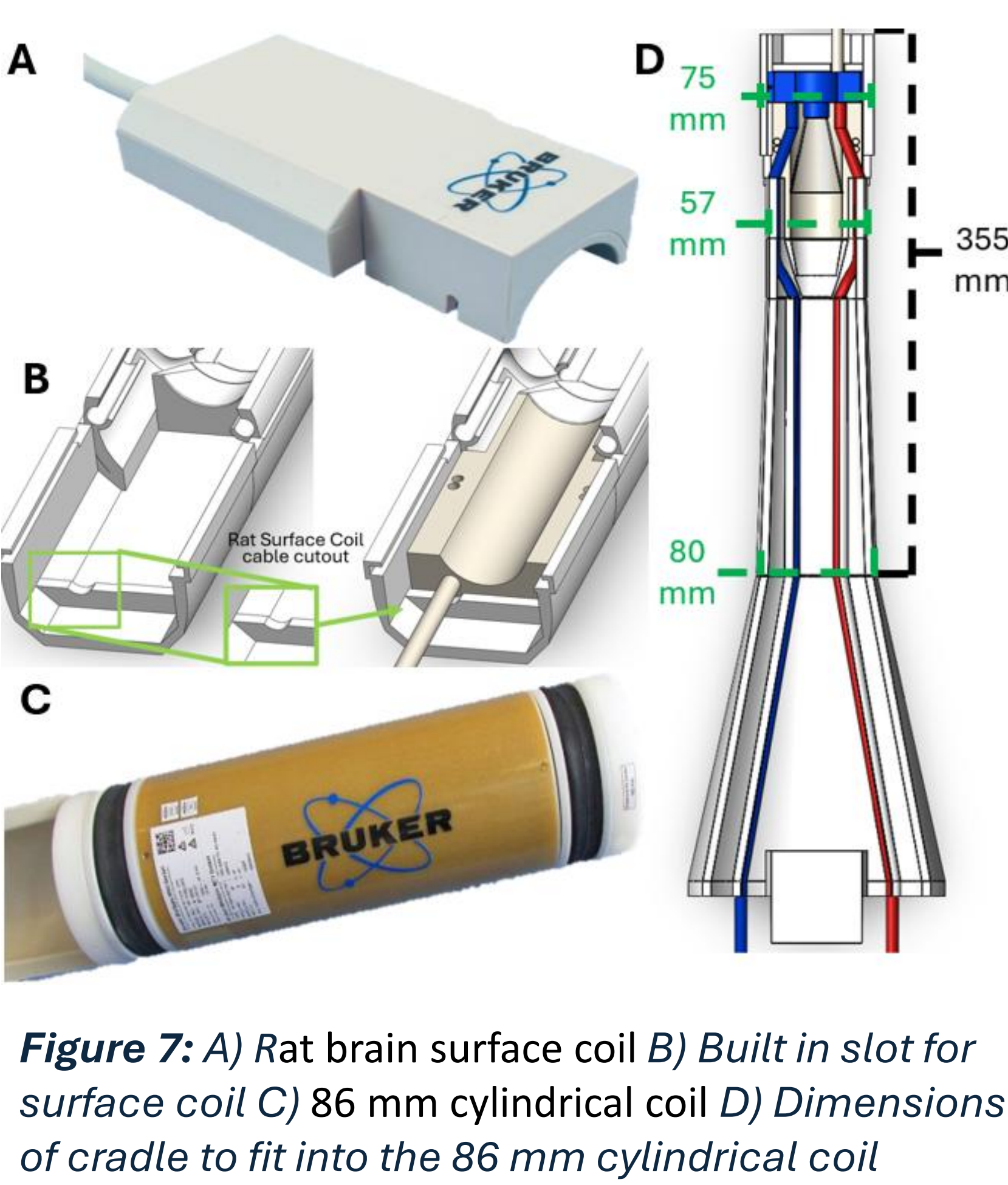


Results – Design Features

Size requirements	
Requirements	Design features
Largest male ferret dimensions	Cradle Dimensions
Body length: 490mm	Length: 500 mm
Head width: 60 mm	Width of head section: 57mm
Body width: 160mm	Width of body section: 160 mm
Adaptability to head size	Anesthesia delivery piece in a rail system See Figure 5:B) &C)
Field of View	Validate with MRI See Figure 8

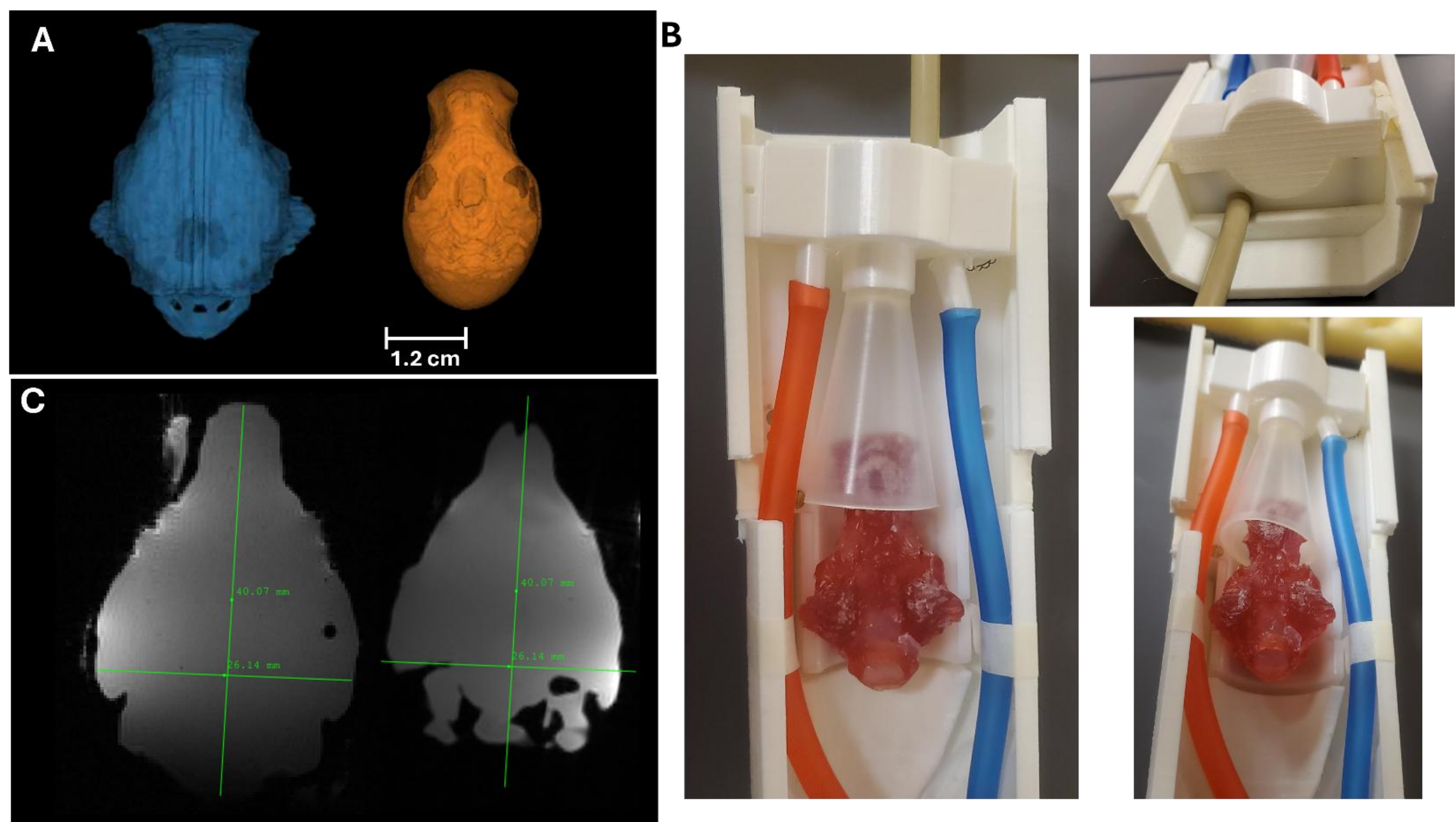


System requirements	
Requirements	Design features
86 mm cylindrical coil and rat brain surface coil used in conjunction	Diameter <80 mm See Figure 7: C) &D) Built in slot See Figure 7: A) & B)
MR compatible material	Poly(lactic acid) (PLA)



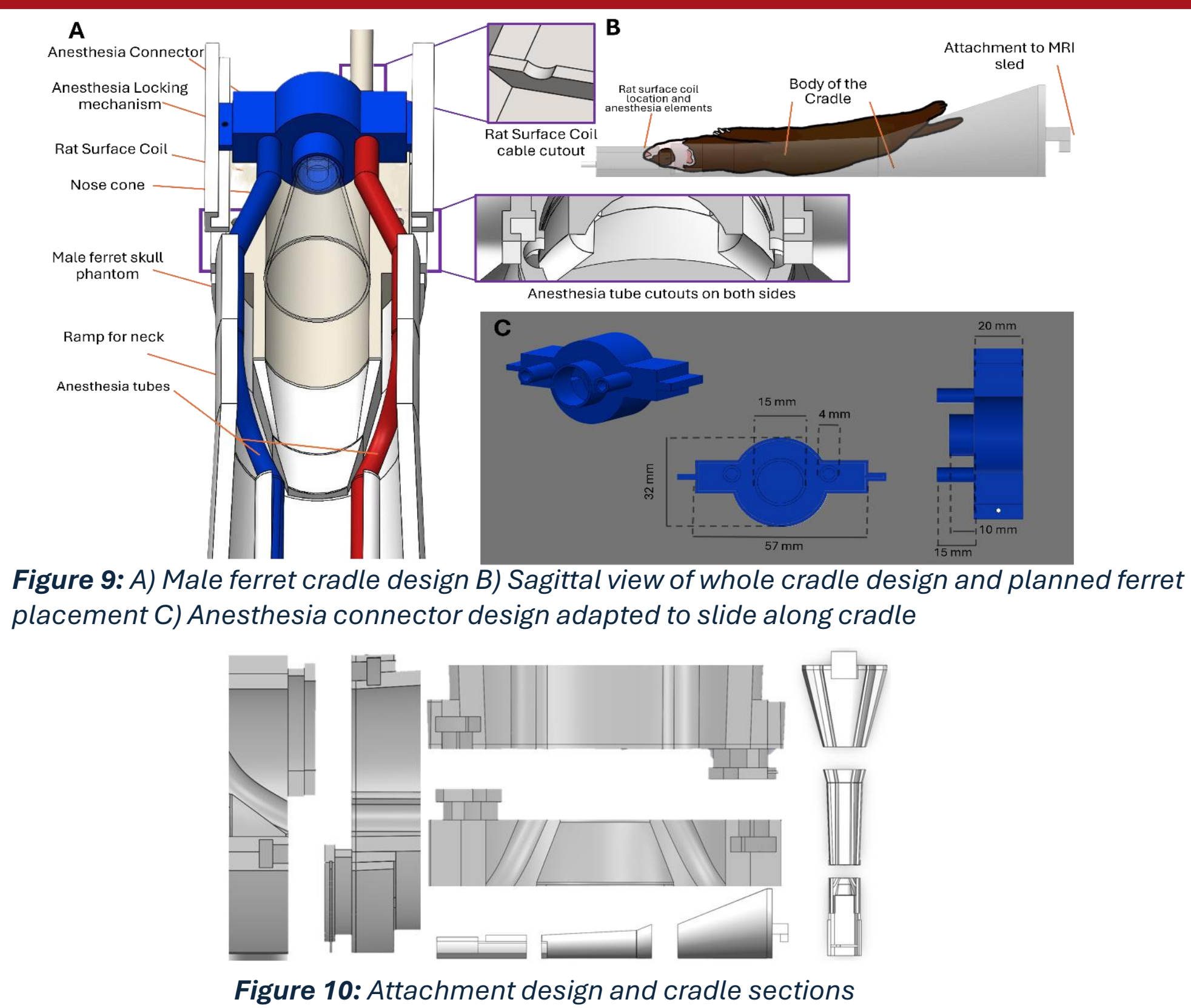
Results – MRI Phantoms Skulls

Phantom testing demonstrated a field of view of 40 x 26 x 14 mm for both the female and male skull, comparable to the in vivo coverage currently achievable for female ferrets. See Figure 8



Conclusions and Future Directions

- Conclusions:**
- Preliminary results demonstrated strong feasibility for in-vivo imaging male ferrets.
 - Features were acceptable by the veterinary staff and performed as expected
- Future directions:**
- Accommodate for sizing changes
 - Design of attachment method between sections
 - Next phase of testing will be to image a male ferret on the new cradle system



References and Website



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Learn more about the MSBIL

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